

Lesson 1

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Tooling setup

The uv package/project manager tool

<https://docs.astral.sh/uv/getting-started/installation/>

Optional uv autocompletion

<https://docs.astral.sh/uv/getting-started/installation/#shell-autocompletion>

Directly run command from package

```
$ uvx cowsay -t "Hello world!"
```

Run Python w/o installing

```
$ uvx python
```

Run the IPython enhanced shell

```
$ uvx ipython
```

Tooling setup, continued...

Installing Python, as opposed to running with uvx

```
$ uv python install --default --preview
```

Run Python when installed

```
$ python3
```

Optionally install other tools

```
$ uv tool install _package_
```

The Python Package Index

<https://pypi.org>

Not covered

- ▶ Legacy tooling, now part of uv: pip/pipx, venv
- ▶ Editor/IDE: Find one you like. Many use VS Code/Codeium

Interactive mode

Invoking the IPython shell/repl

```
$ uvx ipython
```

Using normal commands in IPython, shell convenience only

```
In : !ls
```

...and some Python

```
In : print("Hello world!")
```

```
Out: 'Hello world!'
```

Auto-print convenience, not proper Python

```
In : "Hello world!"
```

```
Out: 'Hello world!'
```

Interactive mode, continued...

Variable assignment

```
In : some_number = 42
```

```
In : some_number
```

```
Out: 42, but only in IPython
```

```
In : type(some_number)
```

```
Out: int
```

Variable naming conventions

<code>my_variable</code>	variables, functions
<code>MyClass</code>	classes
<code>MAX_SIZE</code>	constants

- ▶ More on classes later

Special variables

```
_ = "throw-away variable, by convention"  
_my_var = "this is a private variable"  
__main__ # Special built-in "dunder" methods
```

Non-interactive (real) programming

Script with input/output

```
# script.py  
ip = input("Input an IP address: ")  
print(ip)
```

Run script

```
$ python3 script.py
```

Alternative execution for POSIX systems

Add shebang line on-top

```
#!/usr/bin/env python3  
ip = input("Input an IP address: ")  
print(ip)
```

Make executable

```
$ chmod +x script.py
```

Execute

```
$ ./script.py
```

Comments

```
# This is a comment  
ip_addr = input("Enter an IP address: ")  
print("ip_addr")
```

```
"""
```

```
This is a multiline comment.
```

```
Write your essay here.
```

```
"""
```

Object inspection

```
help(my_var)    # Provides documentation object
```

```
dir(my_var)     # List object members
```

- ▶ In Python, almost everything is an object

Python strings

```
s1 = "string with double quotes"
s2 = '...but single quotes also work'
s3 = """
    This is a
    multiline string.
    Or is it a comment?
    """
```

Some string methods

```
In : s = "banan melon kiwi citron"
```

```
In : s.split()
```

```
Out: ['banan', 'melon', 'kiwi', 'citron']
```

```
In : ip = "192.168.2.1"
```

```
In : parts = ip.split(".")
```

```
In : ".".join(parts)
```

```
Out: '192.168.2.1'
```

Chaining methods

```
In : s = "    Some string    "
```

```
In : s.lower()
```

```
Out: '    some string    '
```

```
In : s.lower().strip()
```

```
Out: 'some string'
```

Python f-strings

```
In : name = "Alice"
```

```
In : f"Hello, {name}!"
```

```
Out: 'Hello, Alice!'
```

```
In : a, b = 5, 10
```

```
In : f"The sum of {a} and {b} is {a + b}."
```

```
Out: 'The sum of 5 and 10 is 15.'
```

```
In : pi = 3.14159
```

```
In : f"Pi is {pi:.2f}."
```

```
Out: 'Pi is 3.14.'
```

More on strings

```
In : "ocean" in "Explore the ocean depths"
```

```
Out: True
```

```
In : "Hello" + ", " + "Alice" + "!"
```

```
Out: 'Hello, Alice!'
```

```
In : for c in "Hello":
```

```
    :     print(c)
```

```
H
```

```
e
```

```
l
```

```
l
```

```
o
```